Network Science

Class 5: BA model

Ganesh Bagler

— Adapted from —
Albert-László Barabási
(With Roberta Sinatra)

Introduction

Section 1

Hubs represent the most striking difference between a random and a scale-free network. Their emergence in many real systems raises several fundamental questions:

•Why does the random network model of Erdős and Rényi fail to reproduce the hubs and the power laws observed in many real networks?

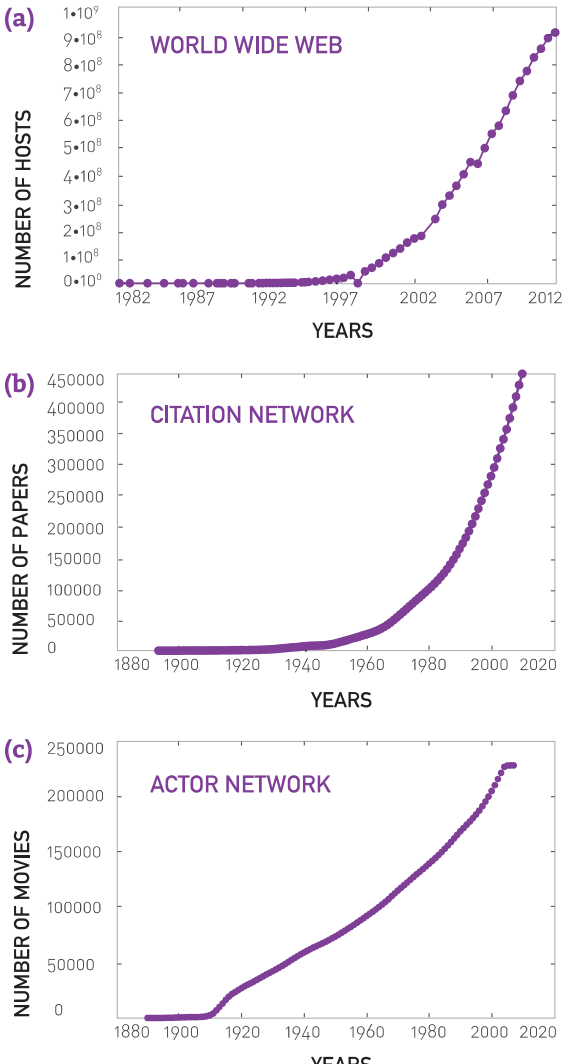
• Why do so different systems as the WWW or the cell converge to a similar scale-free architecture?

Growth and preferential attachment

BA MODEL: Growth

ER model:
the number of nodes, N, is fixed (static models)

networks expand through the addition of new nodes



Barabási & Albert, *Science* **286,** 509 (1999)

BA MODEL: Preferential attachment

ER model: links are added randomly to the network

New nodes prefer to connect to the more connected nodes

Barabási & Albert, *Science* **286,** 509 (1999)

Section 2: Growth and Preferential Attachment

The random network model differs from real networks in two important characteristics:

Growth: While the random network model assumes that the number of nodes is fixed (time invariant), real networks are the result of a growth process that continuously increases.

Preferential Attachment: While nodes in random networks randomly choose their interaction partner, in real networks new nodes prefer to link to the more connected nodes.

Barabási & Albert, *Science* **286,** 509 (1999)

The Barabási-Albert model

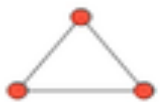
Origin of SF networks: Growth and preferential attachment

(1) Networks continuously expand by the addition of new nodes

WWW : addition of new documents

(2) New nodes prefer to link to highly connected nodes.

WWW : linking to well known sites



Barabási & Albert, *Science* **286,** 509 (1999)

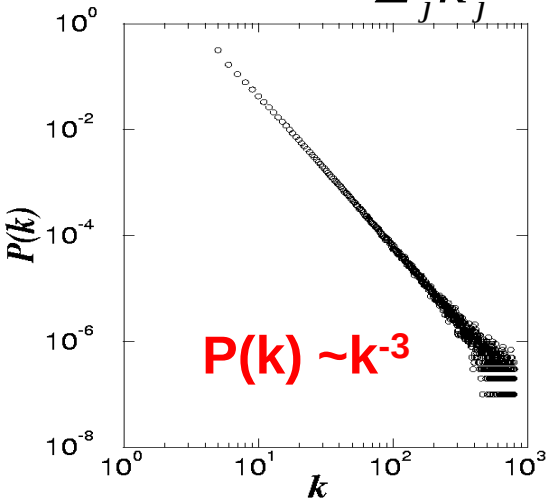
GROWTH:

add a new node with m links

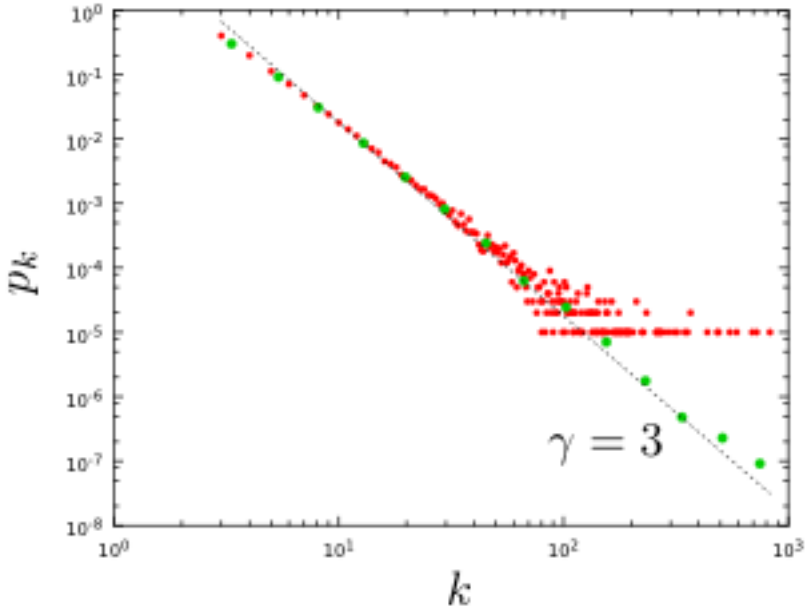
PREFERENTIAL ATTACHMENT:

the probability that a node connects to a node with *k* links is proportional to *k*.

$$\Pi(k_i) = \frac{k_i}{\sum_j k_j}$$



Section 4



György Pólya
PÓLYA PROCESS
MATHEMATICIAN

George Kinsley Zipf
WEALTH DISTRIBUTION
ECONOMIST

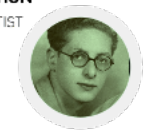
Herbert Alexander Simon
MASTER EQUATION
POLITICAL SCIENTIST

Robert Merton
MATTHEW EFFECT
SOCIOLOGIST

Albert-László Barabási & Réka Albert
PREFERENTIAL ATTACHMENT
NETWORK SCIENTISTS

George Udmy Yule
YULE PROCESS
STATISTICIAN

Robert Gibrat
PROPORTIONAL GROWTH
ECONOMIST



Derek de Solla Price
CUMULATIVE ADVANTAGE
PHYSICIST

MILESTONES

PUBLICATION DATE

1923 1925 1931 1935 1941 1945 1950 1955 1960 1968 1970 1976 1980 1985 1990 1995 1999 2000 2005 2010

XXI

György Pólya (1887-1985)
Preferential attachment made its first appearance in 1923 in the celebrated urn model of the Hungarian mathematician György Pólya [2]. Hence, in mathematics preferential attachment is often called a **Pólya process**.

George Udmy Yule (1871-1951)
used preferential attachment to explain the power-law distribution of the number of species per genus of flowering plants [3]. Hence, in statistics preferential attachment is often called a **Yule process**.

Robert Gibrat (1904-1980)
proposed that the size and the growth rate of a firm are independent. Hence, larger firms grow faster [4]. Called **proportional growth**, this is a form of preferential attachment.

George Kinsley Zipf (1902-1950)
used preferential attachment to explain the fat tailed distribution of wealth in the society [5].

Herbert Alexander Simon (1916-2001)
used preferential attachment to explain the fat-tailed nature of the distributions describing city sizes, word frequencies, or the number of papers published by scientists [6].

Derek de Solla Price (1922-1983)
used preferential attachment to explain the citation statistics of scientific publications, referring to it as **cumulative advantage** [7].

Robert Merton (1910-2003)
In sociology preferential attachment is often called the **Matthew effect**, named by Merton [8] after a passage in the Gospel of Matthew.

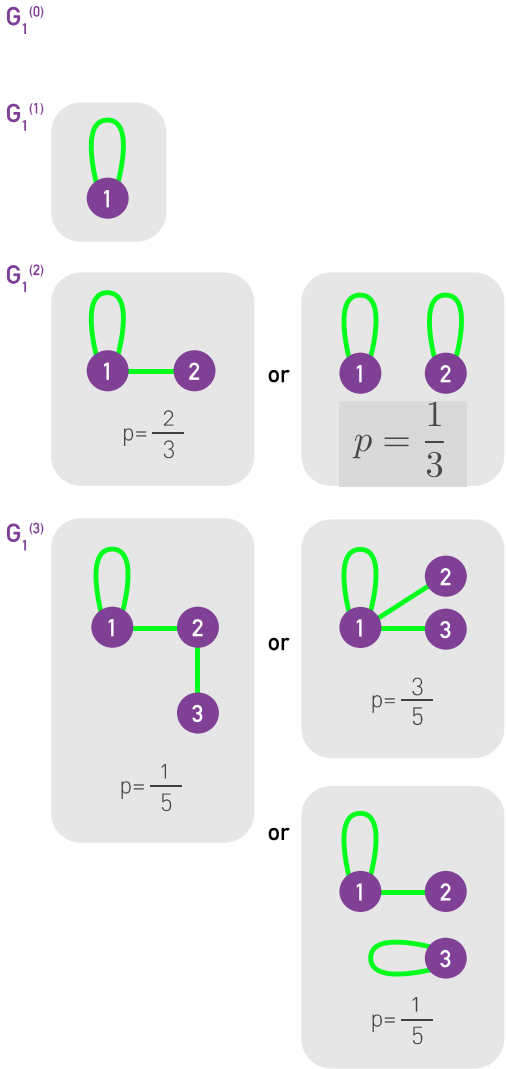
Barabási (1967) & **Albert** (1972)
introduce the term **preferential attachment** in the context of networks [1] to explain the origin of their power-law degree distribution.

Section 4 Linearized Chord Diagram

The definition of the Barabási-Albert model leaves many mathematical details open:

- It does not specify the precise initial configuration of the first m_0 nodes.
- It does not specify whether the m links assigned to a new node are added one by one, or simultaneously. This leads to potential mathematical conflicts: If the links are truly independent, they could connect to the same node i, leading to multi-links.

$$p(i=s)=\begin{cases} \dfrac{k_i}{2t-1} & \text{if } 1 \le s \le t-1 \\ \dfrac{1}{2t-1}, & \text{if } s=t \end{cases}$$



Degree distribution

Degree distribution

$$k_i(t) = m\left(\frac{t}{t_i}\right)^{b} \qquad b = \frac{1}{2}$$

A node *i* can come with equal probability any time between $t_i = m_0$ and t, hence:

$$P(t_i) = \frac{1}{m_0 + t} \qquad P(t_i < t) = \frac{1}{m_0 + t}\int_0^t dt_i = \frac{t}{m_0 + t}$$

$$P(k) = P\left(t_i \le \frac{m^{1/\beta} t}{k^{1/\beta}}\right) = 1 - \frac{m^{1/\beta} t}{k^{1/\beta}(t + m_0)}$$

$$\backslash \quad P(k) = \frac{\partial P(k_i(t) < k)}{\partial k} = \frac{2m^2 t}{m_o + t}\frac{1}{k^3} \sim k^{-\gamma}$$

$$\boxed{\gamma = 3}$$

A.-L.Barabási, R. Albert and H. Jeong, *Physica* A **272,** 173 (1999)

Degree distribution

$$k_i(t) = m\left(\frac{t}{t_i}\right)^b \qquad b = \frac{1}{2}$$

$$P(k) = \frac{2m^2 t}{m_o + t}\frac{1}{k^3} \sim k^{-\gamma}$$

$$\boxed{\gamma = 3}$$

(i) The degree exponent is independent of m.

(ii) As the power-law describes systems of rather different ages and sizes, it is expected that a correct model should provide a time-independent degree distribution. Indeed, asymptotically the degree distribution of the BA model is independent of time (and of the system size N)
→ the network reaches a stationary scale-free state.

(iii) The coefficient of the power-law distribution is proportional to m^2.

A.-L.Barabási, R. Albert and H. Jeong, *Physica* A **272,** 173 (1999)

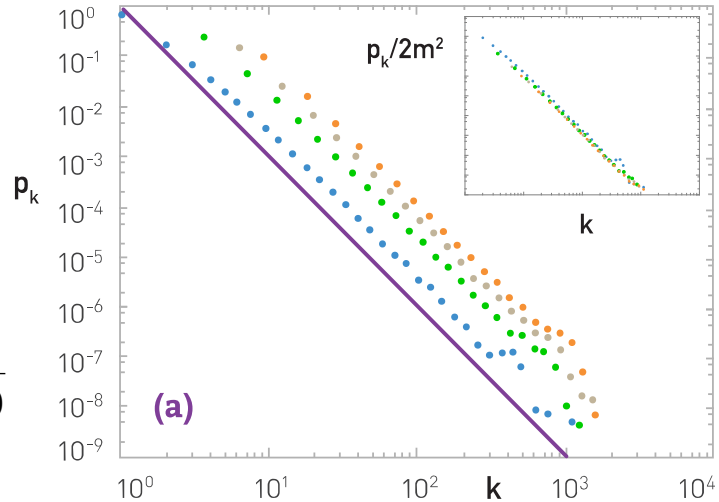
The mean field theory offers the correct scaling, BUT it provides the wrong coefficient of the degree distribution.

So assymptotically it is correct ($k \rightarrow \infty$), but not correct in details (particularly for small k).

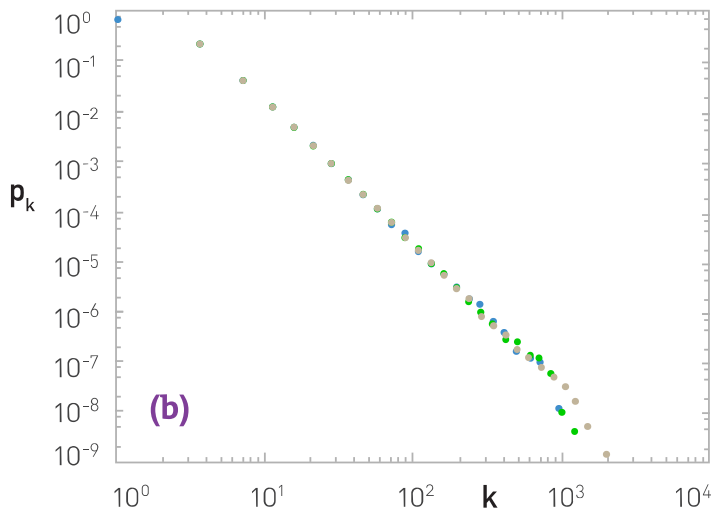
To fix it, we need to calculate P(k) exactly, which we will do next using a rate equation based approach.

NUMERICAL SIMULATION OF THE BA MODEL

$$P(k) = \frac{2m(m+1)}{k(k+1)(k+2)}$$



(a) We generated networks with N=100,000 and m_0=m=1 (blue), 3 (green), 5 (grey), and 7 (orange). The fact that the curves are parallel to each other indicates that γ is independent of m and m_0. The slope of the purple line is -3, corresponding to the predicted degree exponent γ=3. Inset: **(5.11)** predicts p_k~$2m^2$, hence $p_k/2m^2$ should be independent of m. Indeed, by plotting $p_k/2m^2$ vs. k, the data points shown in the main plot collapse into a single curve.



(b) The Barabási-Albert model predicts that p_k is independent of N. To test this we plot p_k for N = 50,000 (blue), 100,000 (green), and 200,000 (grey), with m_0=m=3. The obtained p_k are practically indistinguishable, indicating that the degree distribution is stationary, i.e. independent of time and system size.

absence of growth and preferential attachment

MODEL A

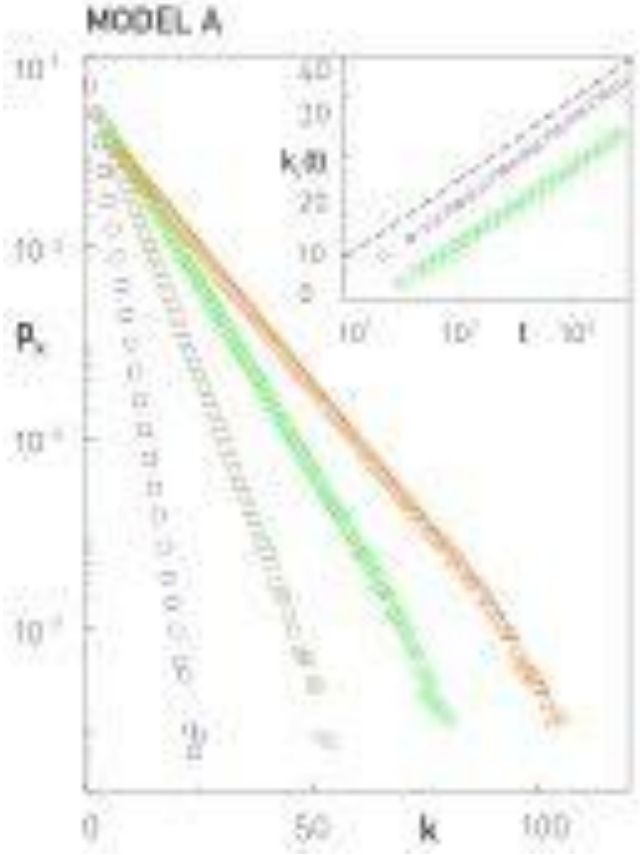
growth ~~preferential attachment~~

$\Pi(k_i)$: uniform

$$\frac{\P k_i}{\P t} = A\mathrm{P}(k_i) = \frac{m}{m_0 + t - 1}$$

$$k_i(t) = m\ln\left(\frac{m_0 + t - 1}{m + t_i - 1}\right) + m$$

$$P(k) = \frac{e}{m}\exp\left(-\frac{k}{m}\right) \sim e^{-k}$$

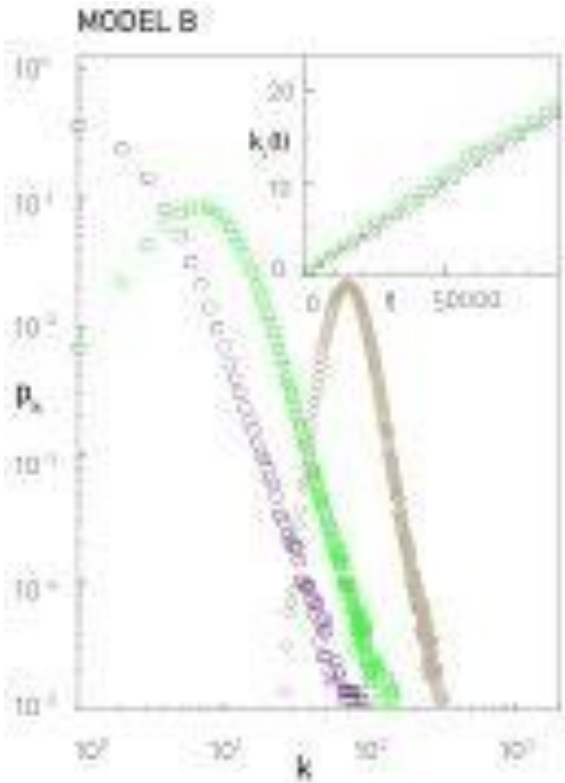


MODEL B

~~growth~~ preferential attachment

$$\frac{\partial k_i}{\partial t} = A\Pi(k_i) + \frac{1}{N} = \frac{N}{N-1}\frac{k_i}{2t} + \frac{1}{N}$$

$$k_i(t) = \frac{2(N-1)}{N(N-2)}t + Ct^{\frac{N}{2(N-1)}} \sim \frac{2}{N}t$$



p_k : power law (initially) →

→ Gaussian → Fully Connected

Do we need both growth and preferential attachment?

YEP.

Measuring preferential attachment

Section 7 Measuring preferential attachment

$$\frac{\partial k_i}{\partial t} \propto \Pi(k_i) \sim \frac{\Delta k_i}{\Delta t}$$

Plot the change in the degree Δk during a fixed time Δt for nodes with degree k.

To reduce noise, plot the integral of $\Pi(k)$ over k:

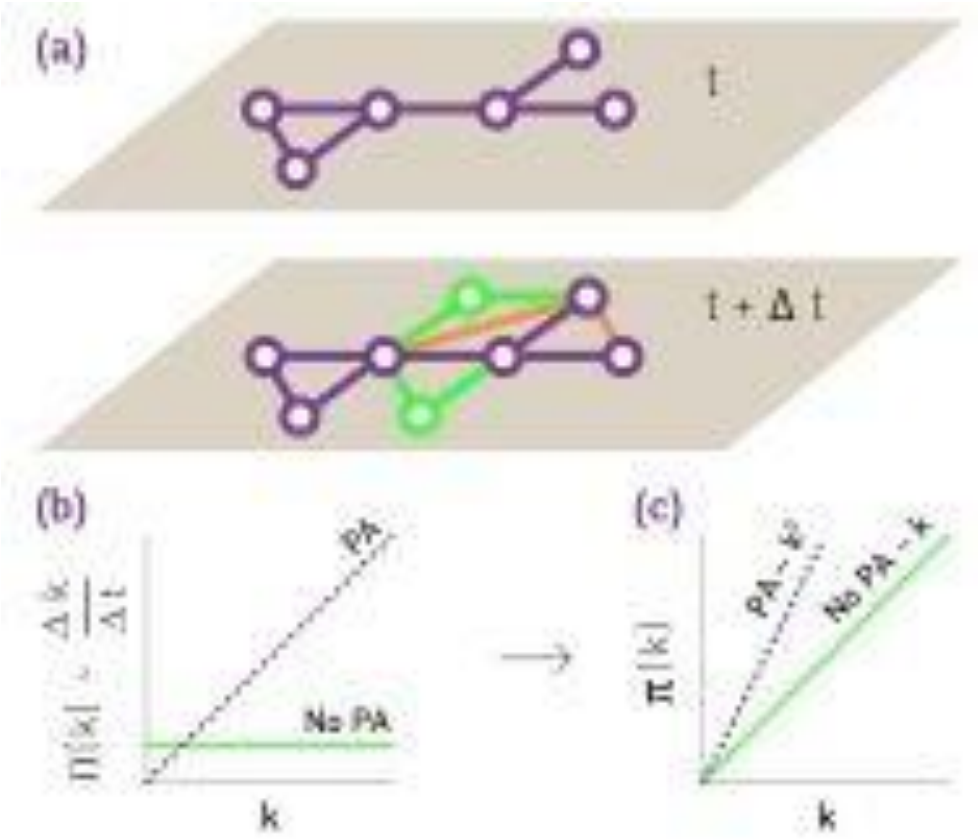
$$\kappa(k) = \sum_{K<k} \Pi(K)$$

No pref. attach:

$\kappa \sim k$ ———

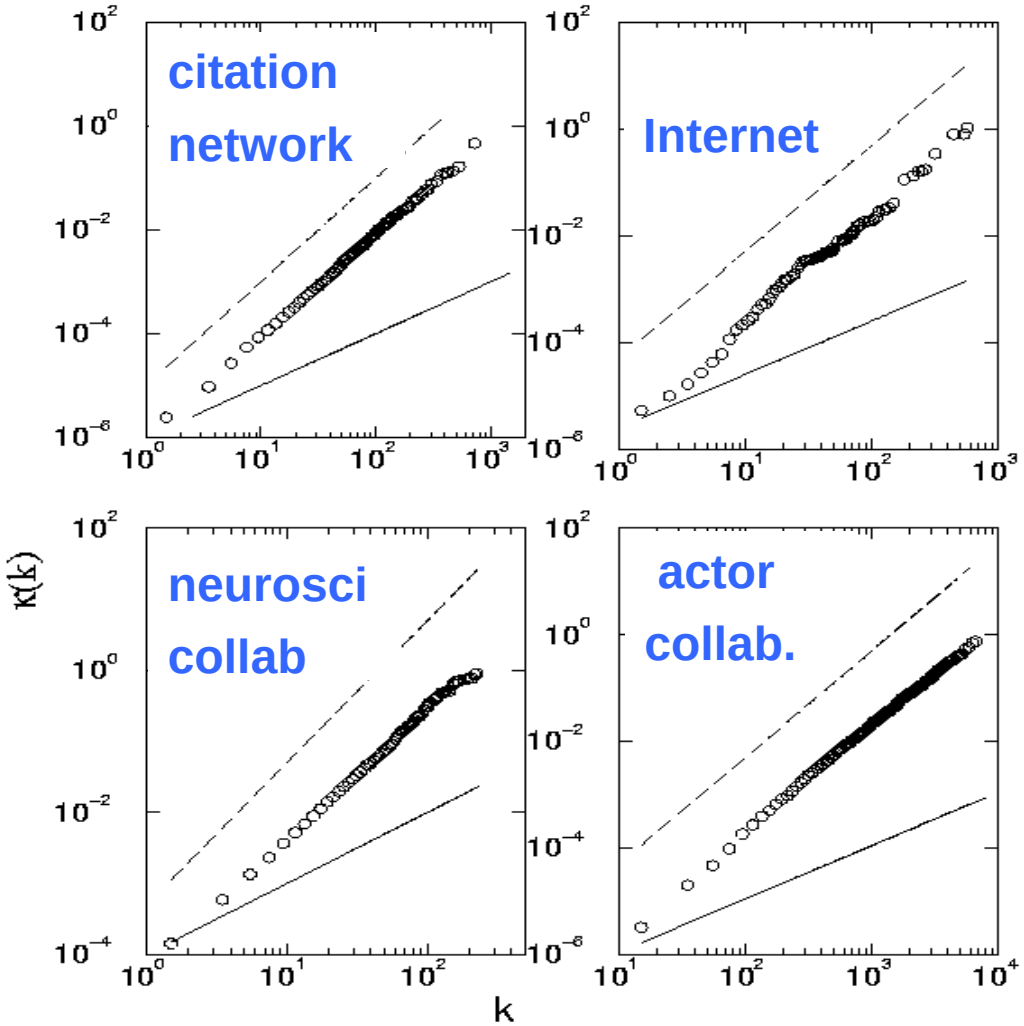
Linear pref. attach:

$\kappa \sim k^2$ - - - -



(Jeong, Neda, A.-L. B, Europhys Letter 2003; cond-mat/0104131)

Measuring preferential attachment



Plots shows the integral of Π(k) over k:

$$\kappa(k) = \sum_{K<k} \Pi(K)$$

No pref. attach:

$\kappa \sim k$ ———

Linear pref. attach:

$\kappa \sim k^2$ - - - -

$$\Pi(k) \approx A + k^{\alpha}, \ \alpha \leq 1$$

The origins of preferential attachment

Link selection model

Link selection model -- perhaps the simplest example of a local or random mechanism capable of generating preferential attachment.

***Growth**: at each time step we add a new node to the network.*

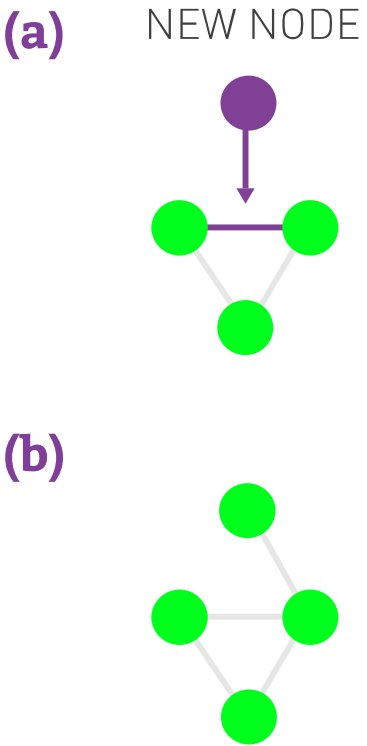
***Link selection**: we select a link at random and connect the new node to one of nodes at the two ends of the selected link.*

To show that this simple mechanism generates linear preferential attachment, we write the probability that the node at the end of a randomly chosen link has degree k as

$$q_k = C k p_k$$

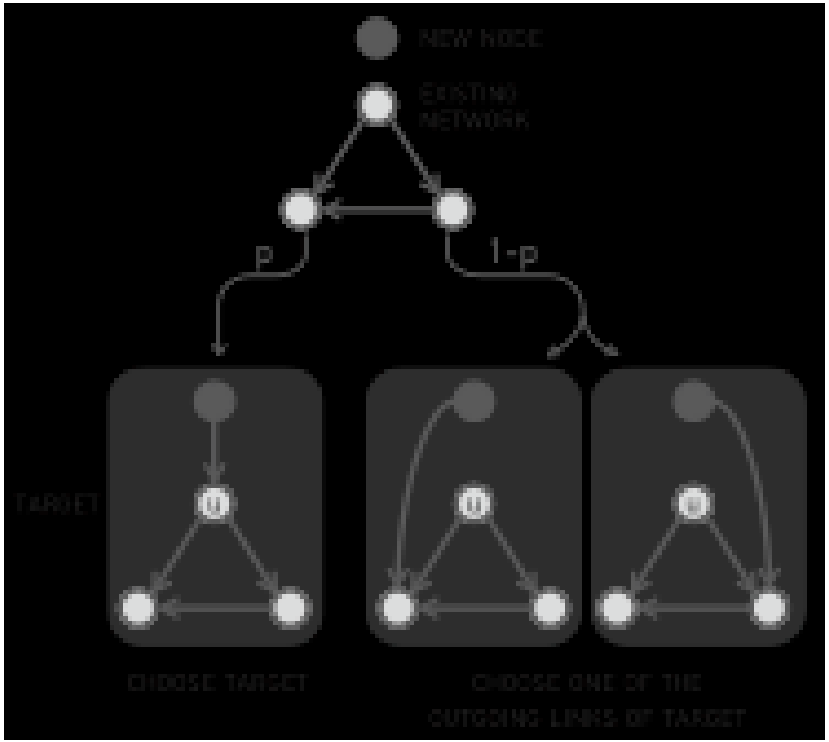
In (5.26) C can be calculated using the normalization condition $\Sigma q_k = 1$, obtaining $C=1/\langle k \rangle$. Hence the probability to find a degree-k node at the end of a randomly chosen link is

$$q_k = \frac{k p_k}{\langle k \rangle}, \qquad (5.27)$$



Section 9

Copying model



(a) Random Connection: *with probability p the new node links to u.*

(b) Copying*: with probability we randomly choose an outgoing link of node u and connect the new node to the selected link's target. Hence the new node "copies" one of the links of an earlier node*

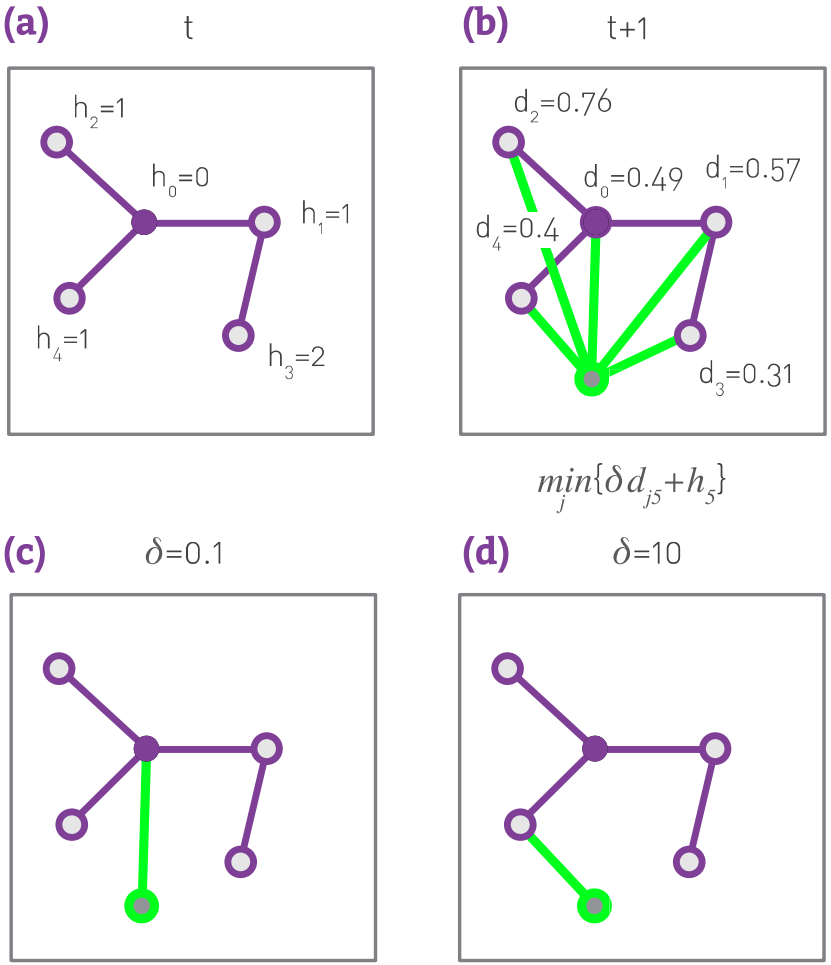
(a) the probability of selecting a node is 1/N.
(b) is equivalent with selecting a node linked to a randomly selected link. The probability of selecting a degree-k node through the copying process of step (b) is k/2L for undirected networks.
The likelihood that the new node will connect to a degree-k node follows preferential attachment

$$\Pi(k) = p/N + (1-p)k/(2L)$$

Social networks: *Copy your friend's friends.*
Citation Networks*: Copy references from papers we read.*
Protein interaction networks*: gene duplication,*

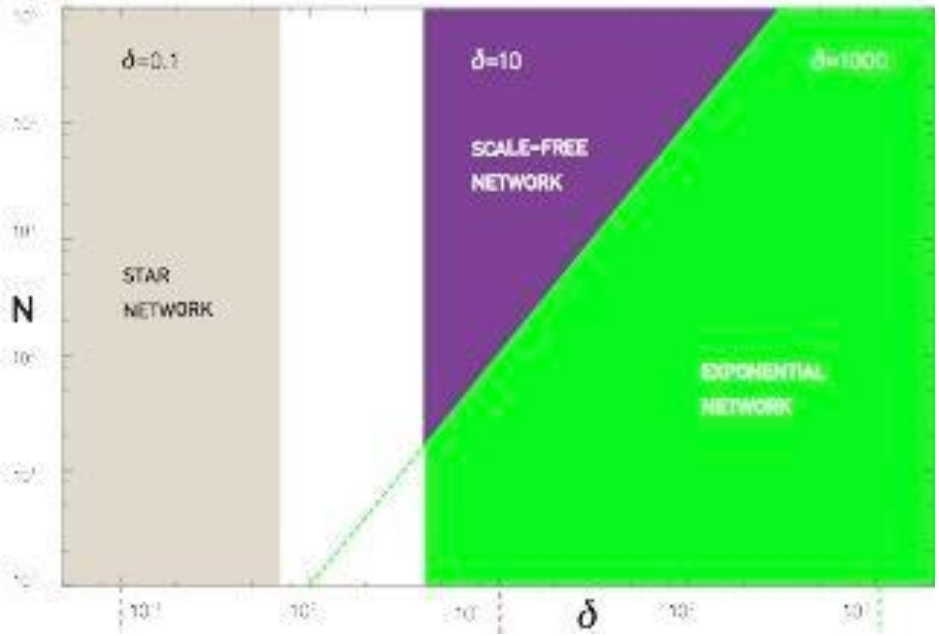
Optimization model

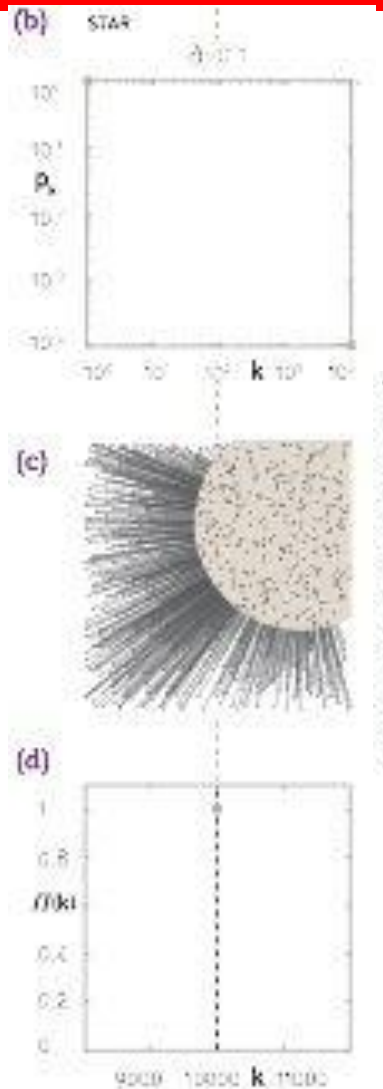
$$C_i = \min_j[\delta d_{ij} + h_j]$$



Optimization model

$$C_i = \min_j[\delta d_{ij} + h_j]$$

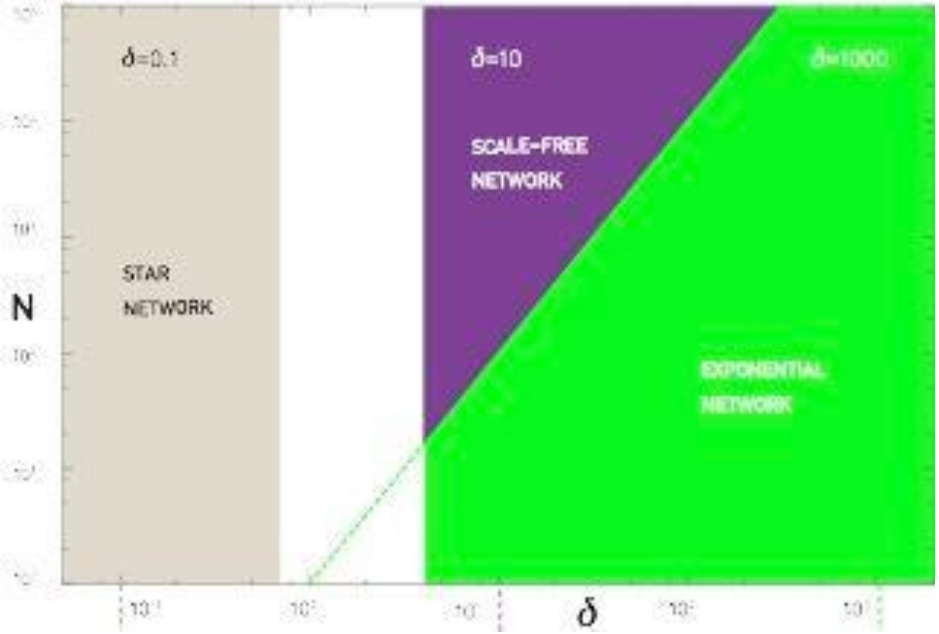


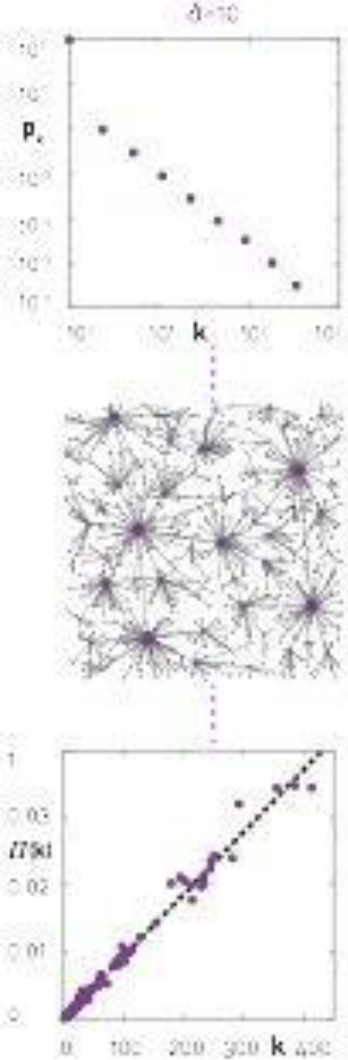


The vertical boundary of the star configuration is at $\delta=(1/2)^{1/2}$. This is the inverse of the maximum distance between two nodes on a square lattice with unit length, over which the model is defined. Therefore if $\delta < (1/2)^{1/2}$, for any new node $\delta d_{i0} < 1$ and the cost (5.26) of connecting to the central node is $C_i = \delta d_{i0} + 0$, always lower than connecting to any other node at the cost of $f(i,j) = \delta d_{ij} + 1$. Therefore for $\delta < (1/2)^{1/2}$ all nodes connect to node 0, resulting in a network dominated by a single hub (star-and-spoke network (c)).

Optimization model

$$C_i = \min_j[\delta d_{ij} + h_j]$$

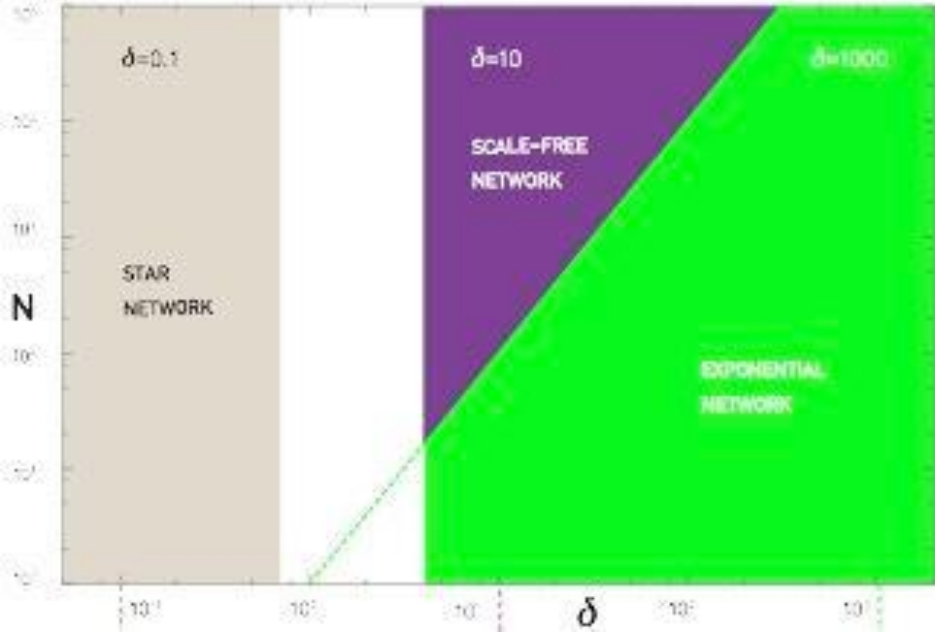


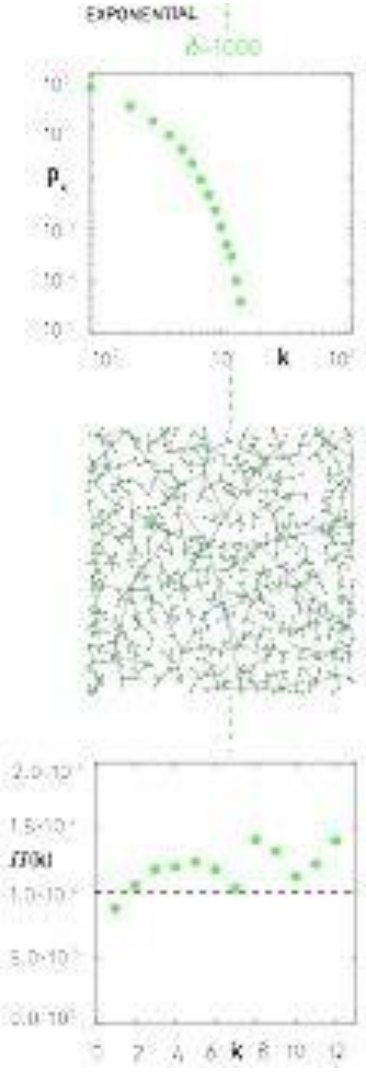


The oblique boundary of the scale-free regime is $\delta \sim N^{1/2}$. Indeed, if nodes are placed randomly on the unit square, then the typical distance between neighbors decreases as $N^{-1/2}$. Hence, if $d \sim N^{-1/2}$ then $\delta d \gtrsim h_j$ for most node pairs. Typically the path length to the central node h_j grows slower than N (in small-world networks $h_j \sim \log N$, in scale-free networks $h_j \sim \ln(\ln N)$). Therefore C_i is dominated by the δd_{ij} term and the smallest C_i is achieved by minimizing the distance-dependent term. Note that strictly speaking the transition only occurs in the $N \to \infty$ limit.

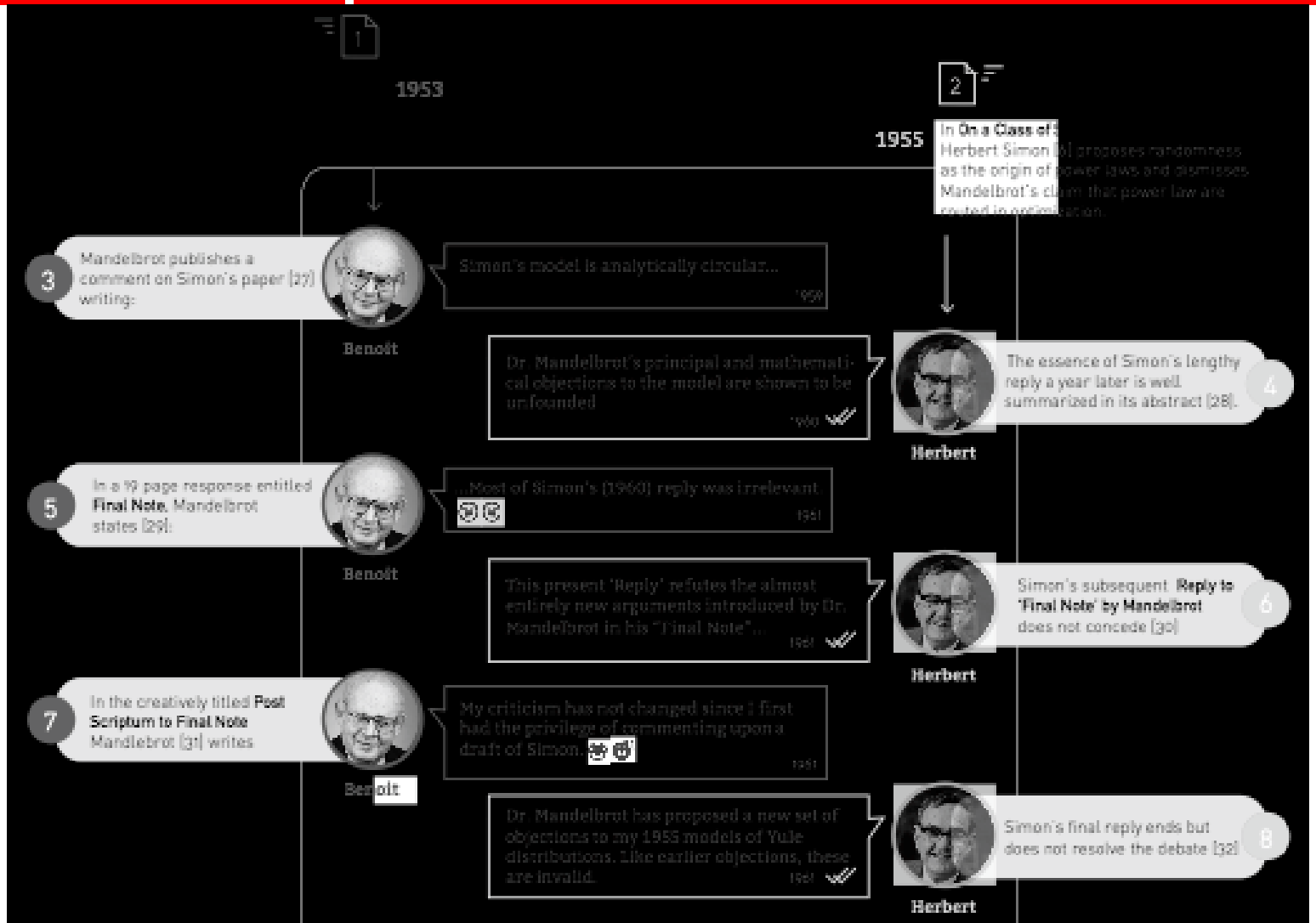
Optimization model

$$C_i = \min_j[\delta d_{ij} + h_j]$$





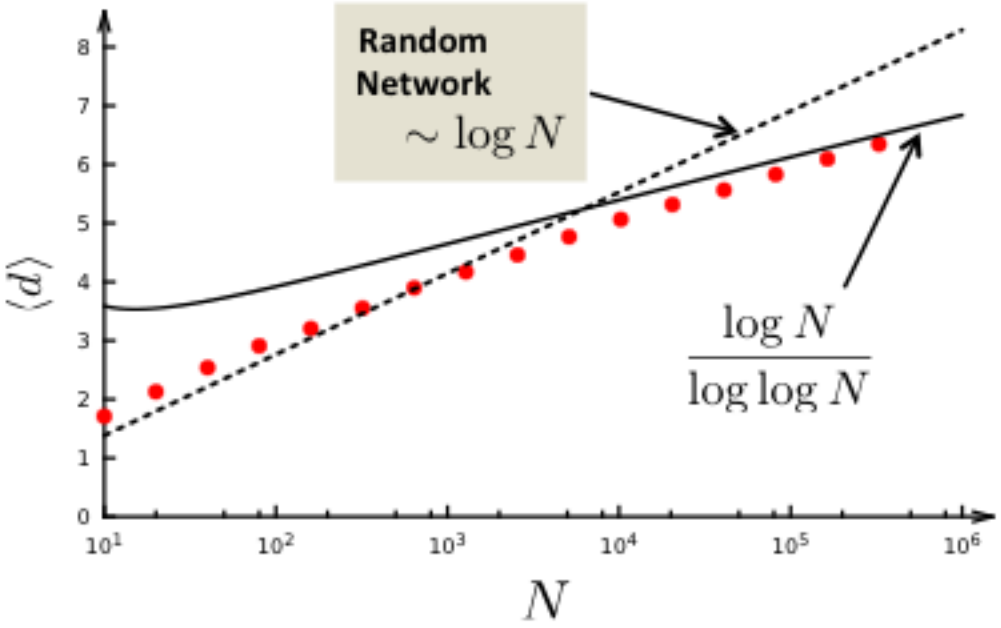
Section 9


1
1953
2
1955
In On a Class of: Herbert Simon [6] proposes randomness as the origin of power laws and dismisses Mandelbrot's claim that power law are rooted in optimization.
3
Mandelbrot publishes a comment on Simon's paper [27] writing:
Simon's model is analytically circular...
1959
Benoit
Dr. Mandelbrot's principal and mathematical objections to the model are shown to be unfounded
1960
Herbert
The essence of Simon's lengthy reply a year later is well summarized in its abstract [28].
4
5
In a 19 page response entitled Final Note, Mandelbrot states [29]:
...Most of Simon's (1960) reply was irrelevant.
1961
Benoit
This present 'Reply' refutes the almost entirely new arguments introduced by Dr. Mandelbrot in his "Final Note"...
1961
Herbert
Simon's subsequent Reply to 'Final Note' by Mandelbrot does not concede [30]
6
7
In the creatively titled Post Scriptum to Final Note Mandelbrot [31] writes
My criticism has not changed since I first had the privilege of commenting upon a draft of Simon's
1961
Benoit
Dr. Mandelbrot has proposed a new set of objections to my 1955 models of Yule distributions. Like earlier objections, these are invalid.
1961
Herbert
Simon's final reply ends but does not resolve the debate [32]
8

Diameter and clustering coefficient

Diameter

$$D \sim \frac{\log N}{\log\log N}$$

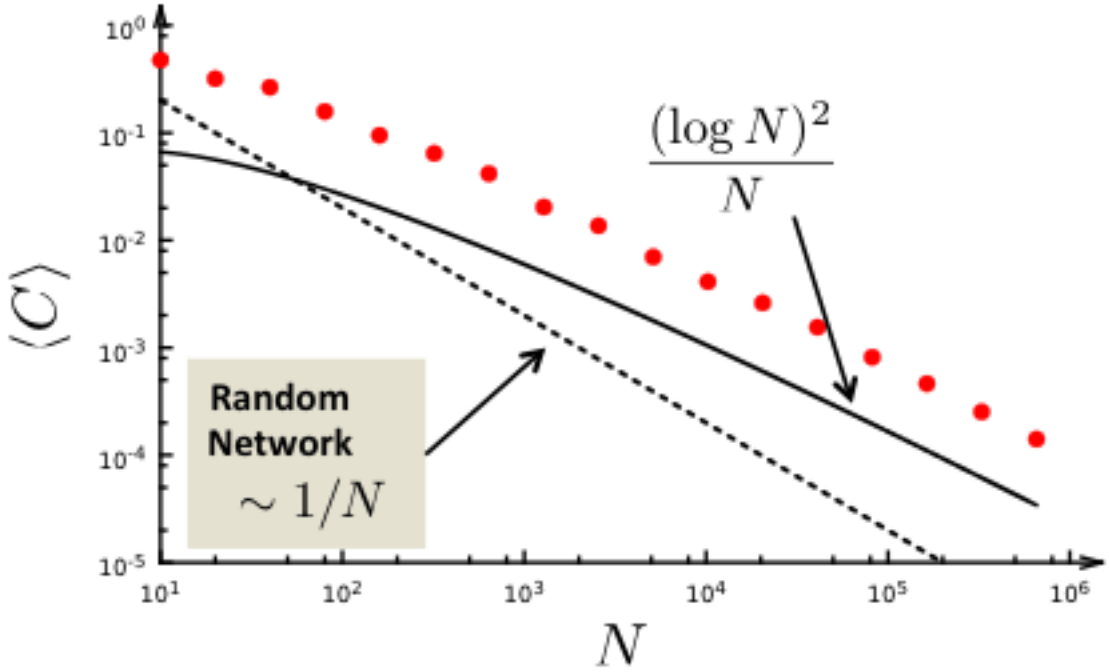


Bollobas, Riordan, 2002

Reminder: for a random graph we have:

$$C_{rand} = \frac{<k>}{N} \sim N^{-1}$$

What is the functional form of C(N)?

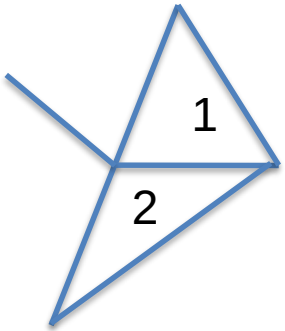


$$C = \frac{m}{8}\frac{(\ln N)^2}{N}$$

Konstantin Klemm, Victor M. Eguiluz,
Growing scale-free networks with small-world behavior,
Phys. Rev. E 65, 057102 (2002), cond-mat/0107607

CLUSTERING COEFFICIENT OF THE BA MODEL

$$C = \frac{Nr(\square)}{\frac{k(k-1)}{2}}$$



$$C = \frac{2}{6}$$

Denote the probability to have a link between node *i* and *j* with *P(i,j)*
The probability that three nodes *i,j,l* form a triangle is *P(i,j)P(i,l)P(j,l)*

The expected number of triangles in which a node *l* with degree k_l participates is thus:

$$Nr_l(\square) = \sum_{i=1}^{N} di \sum_{j=1}^{N} dj P(i,j)P(i,l)P(j,l)$$

We need to calculate P(i,j).

CLUSTERING COEFFICIENT OF THE BA MODEL

Calculate P(i,j).

Node j arrives at time $t_j=j$ and the probability that it will link to node i with degree k_i already in the network is determined by preferential attachment:

$$P(i,j) = m\Pi(k_i(j)) = m\frac{k_i(j)}{\sum_{l=1}^{j} k_l} = m\frac{k_i(j)}{2mj}$$

$$k_i(t) = m\left(\frac{t}{t_i}\right)^{1/2} = m\left(\frac{j}{i}\right)^{1/2}$$

Where we used that the arrival time of node j is $t_j=j$ and the arrival time of node is $t_i=i$

$$P(i,j) = \frac{m}{2}(ij)^{-\frac{1}{2}}$$

$$Nr_l(\square) = \int_{i=1}^{N} di \int_{j=1}^{N} dj P(i,j)P(i,l)P(j,l) = \frac{m^3}{8}\int_{i=1}^{N} di \int_{j=1}^{N} dj (ij)^{-\frac{1}{2}}(il)^{-\frac{1}{2}}(jl)^{-\frac{1}{2}} = \frac{m^3}{8l}\int_{i=1}^{N}\frac{di}{i}\int_{j=1}^{N}\frac{dj}{j} = \frac{m^3}{8l}(\ln N)^2$$

$$C = \frac{\frac{m^3}{8l}(\ln N)^2}{k_l(k_l - 1)/2}$$

$$k_l(t) = m\left(\frac{N}{l}\right)^{1/2}$$

Which is the degree of node l at current time, at time $t=N$

Let us approximate:

$$k_l(k_l - 1) \approx k_l^2 = m^2\frac{N}{l}$$

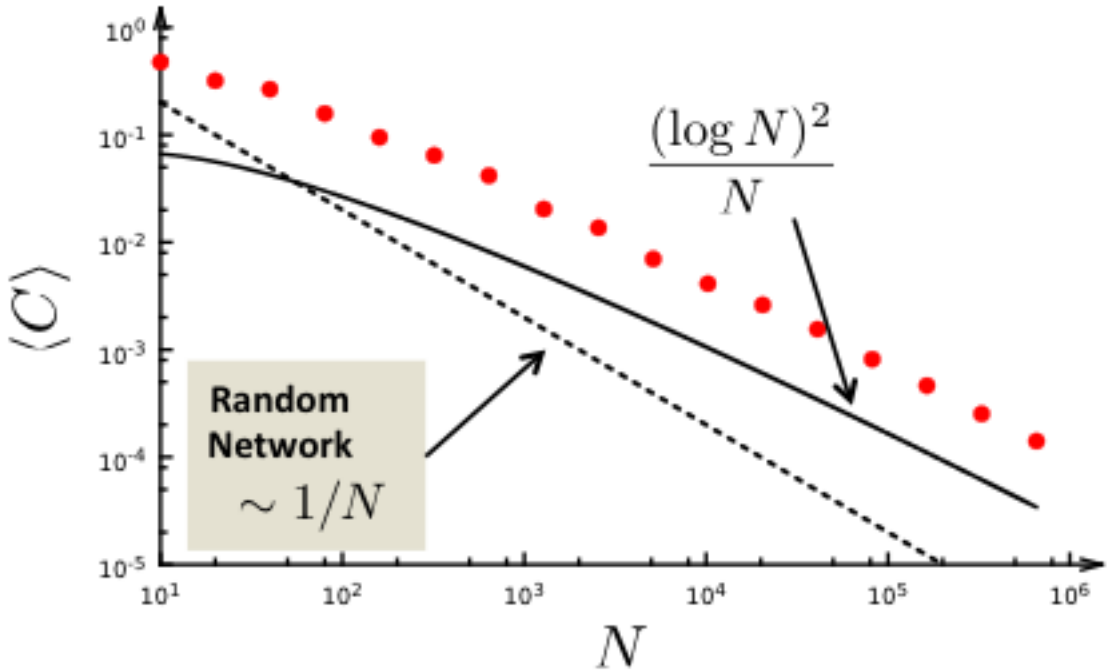
$$C = \frac{m}{8}\frac{(\ln N)^2}{N}$$

There is a factor of two difference... Where does it come from?

Reminder: for a random graph we have:

$$C_{rand} = \frac{<k>}{N} \sim N^{-1}$$

What is the functional form of C(N)?



$$C = \frac{m}{8}\frac{(\ln N)^2}{N}$$

Konstantin Klemm, Victor M. Eguiluz,
Growing scale-free networks with small-world behavior,
Phys. Rev. E 65, 057102 (2002), cond-mat/0107607

Section 11: Summary

Number of Nodes
$N = t$

Number of Links
$N = mt$

Average Degree
$\langle k \rangle = 2m$

Degree Dynamics
$k_i(t) = m\,(t/t_i)^{\beta}$

Dynamical Exponent
$\beta = 1/2$

Degree Distribution
$p_k \sim k^{\gamma}$

Degree Exponent
$\gamma = 3$

Average Distance
$\langle d \rangle \sim \log N / \log \log N$

Clustering Coefficient
$\langle C \rangle \sim (\ln N)^2 / N$

Consequently, the modeling philosophy behind the model is simple: *to understand the topology of a complex system, we need to describe how it came into being.*

The network grows, but the degree distribution is stationary.

Section 11: Summary

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- The model predicts γ=3 while the degree exponent of real networks varies between 2 and 5 (Table 4.2).

- Many networks, like the WWW or citation networks, are directed, while the model generates undirected networks.

- Many processes observed in networks, from linking to already existing nodes to the disappearance of links and nodes, are absent from the model.

- The model does not allow us to distinguish between nodes based on some intrinsic characteristics, like the novelty of a research paper or the utility of a webpage.

- While the Barabási-Albert model is occasionally used as a model of the Internet or the cell, in reality it is not designed to capture the details of any particular real network. It is a minimal, proof of principle model whose main purpose is to capture the basic mechanisms responsible for the emergence of the scale-free property. Therefore, if we want to understand the evolution of systems like the Internet, the cell or the WWW, we need to incorporate the important details that contribute to the time evolution of these systems, like the directed nature of the WWW, the possibility of internal links and node and link removal.